

Program 1: Creating a Maven Project

Step 1: `sudo apt install openjdk-11-jdk -y`

Installs Java Development Kit

(check version installed using `java -version`)

Step 2: `sudo apt install maven -y`

Installs Maven for building Java projects

Step 3: `sudo apt install gradle -y`

Installs Gradle

Step 4: `sudo apt install ansible -y`

Installs ansible for automation management

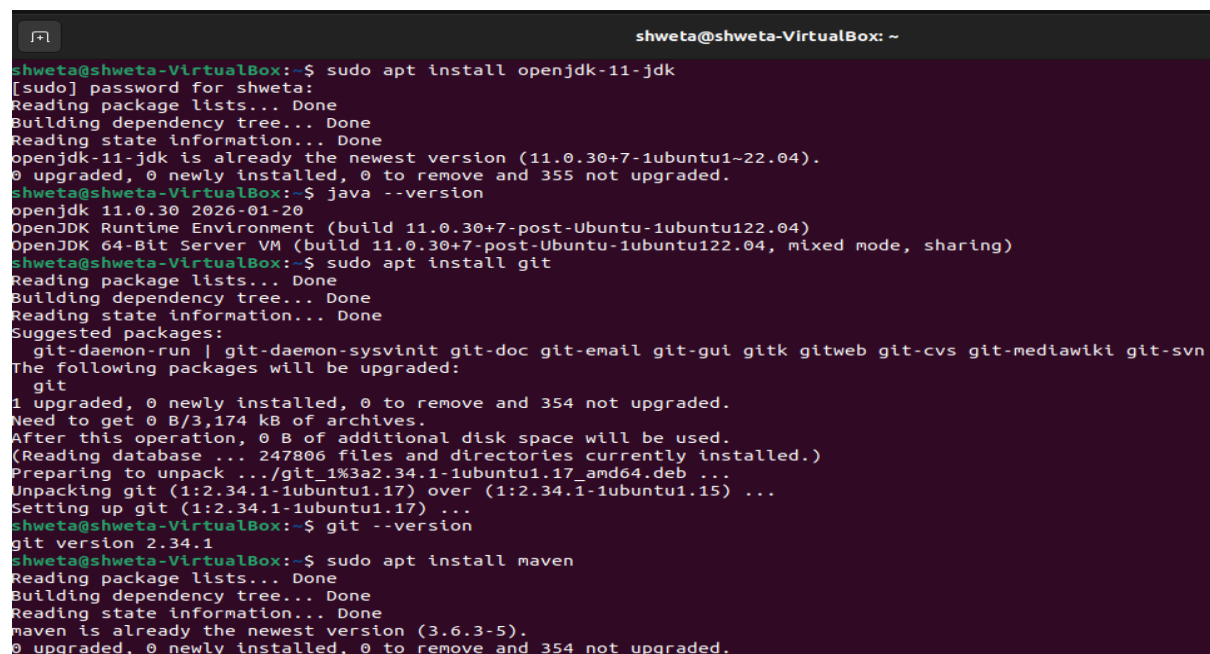
Step 5:

`sudo apt install wget -y`

`wget` https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb

`sudo dpkg -I google-chrome-stable_current_amd64.deb` (Downloading chrome)

`google-chrome -version`



```
shweta@shweta-VirtualBox: ~  
shweta@shweta-VirtualBox:~$ sudo apt install openjdk-11-jdk  
[sudo] password for shweta:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
openjdk-11-jdk is already the newest version (11.0.30+7-1ubuntu1-22.04).  
0 upgraded, 0 newly installed, 0 to remove and 355 not upgraded.  
shweta@shweta-VirtualBox:~$ java --version  
openjdk 11.0.30 2026-01-20  
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu122.04)  
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu122.04, mixed mode, sharing)  
shweta@shweta-VirtualBox:~$ sudo apt install git  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Suggested packages:  
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn  
The following packages will be upgraded:  
  git  
1 upgraded, 0 newly installed, 0 to remove and 354 not upgraded.  
Need to get 0 B/3,174 kB of archives.  
After this operation, 0 B of additional disk space will be used.  
(Reading database ... 247806 files and directories currently installed.)  
Preparing to unpack .../git_1%3a2.34.1-1ubuntu1.17_amd64.deb ...  
Unpacking git (1:2.34.1-1ubuntu1.17) over (1:2.34.1-1ubuntu1.15) ...  
Setting up git (1:2.34.1-1ubuntu1.17) ...  
shweta@shweta-VirtualBox:~$ git --version  
git version 2.34.1  
shweta@shweta-VirtualBox:~$ sudo apt install maven  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
maven is already the newest version (3.6.3-5).  
0 upgraded, 0 newly installed, 0 to remove and 354 not upgraded.
```

Step 6: To create a basic Maven Java project, run the following command:

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp-
DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

Breakdown of the command:

-DgroupId=com.example – project’s base package (reverse-domain style).

-DartifactId=my-maven-app – defines the name of the project (will also be the folder name).

-DarchetypeArtifactId=maven-archetype-quickstart – sets the Maven template to be the basic Java application.

-DinteractiveMode=false – deactivates interactive mode and uses the given values automatically.

```
shweta@shweta-VirtualBox:~$ google-chrome --version
Google Chrome 145.0.7632.116
shweta@shweta-VirtualBox:~$ cd Devops
shweta@shweta-VirtualBox:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/shweta/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp03
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shweta/Devops/MyMavenSeleniumApp03
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 6.013 s
[INFO] Finished at: 2026-03-15T05:10:16+05:30
[INFO]
shweta@shweta-VirtualBox:~/Devops$
```

Step 7: cd MyMavenApp: use this command to move to directory of Maven app we just created

Step 8: gedit pom.xml: use this command to open the pom.xml file using gedit editor.



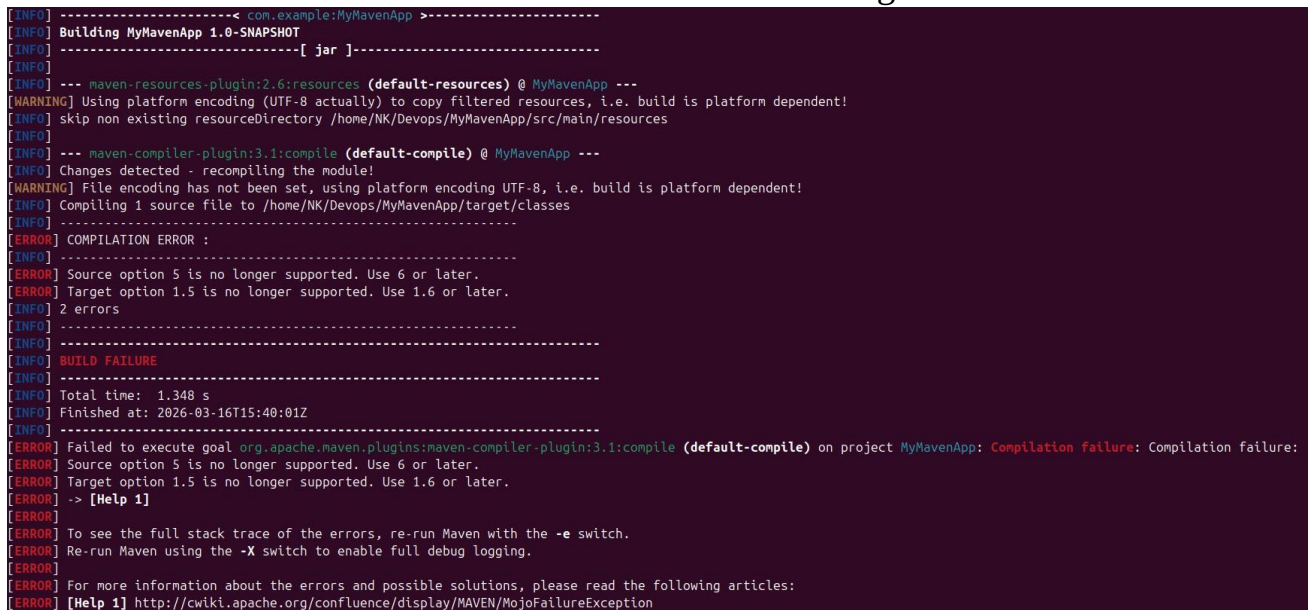
```

1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3   <modelVersion>4.0.0</modelVersion>
4   <groupId>com.example</groupId>
5   <artifactId>MyMavenApp</artifactId>
6   <packaging>jar</packaging>
7   <version>1.0-SNAPSHOT</version>
8   <name>MyMavenApp</name>
9   <url>http://maven.apache.org</url>
10  <dependencies>
11    <dependency>
12      <groupId>junit</groupId>
13      <artifactId>junit</artifactId>
14      <version>3.8.1</version>
15      <scope>test</scope>
16    </dependency>
17  </dependencies>
18 </project>

```

Step 9: mvn compile: it compiles the source code and check for errors

When we run above command for the first time we would get some error like this:



```

[INFO] ----- com.example:MyMavenApp -----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/NK/Devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/NK/Devops/MyMavenApp/target/classes
[INFO]
[ERROR] COMPILATION ERROR :
[INFO]
[ERROR] Source option 5 is no longer supported. Use 6 or later.
[ERROR] Target option 1.5 is no longer supported. Use 1.6 or later.
[INFO] 2 errors
[INFO]
[INFO] BUILD FAILURE
[INFO]
[INFO] Total time: 1.348 s
[INFO] Finished at: 2026-03-16T15:40:01Z
[INFO]
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.1:compile (default-compile) on project MyMavenApp: Compilation failure: Compilation failure:
[ERROR] Source option 5 is no longer supported. Use 6 or later.
[ERROR] Target option 1.5 is no longer supported. Use 1.6 or later.
[ERROR] -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException

```

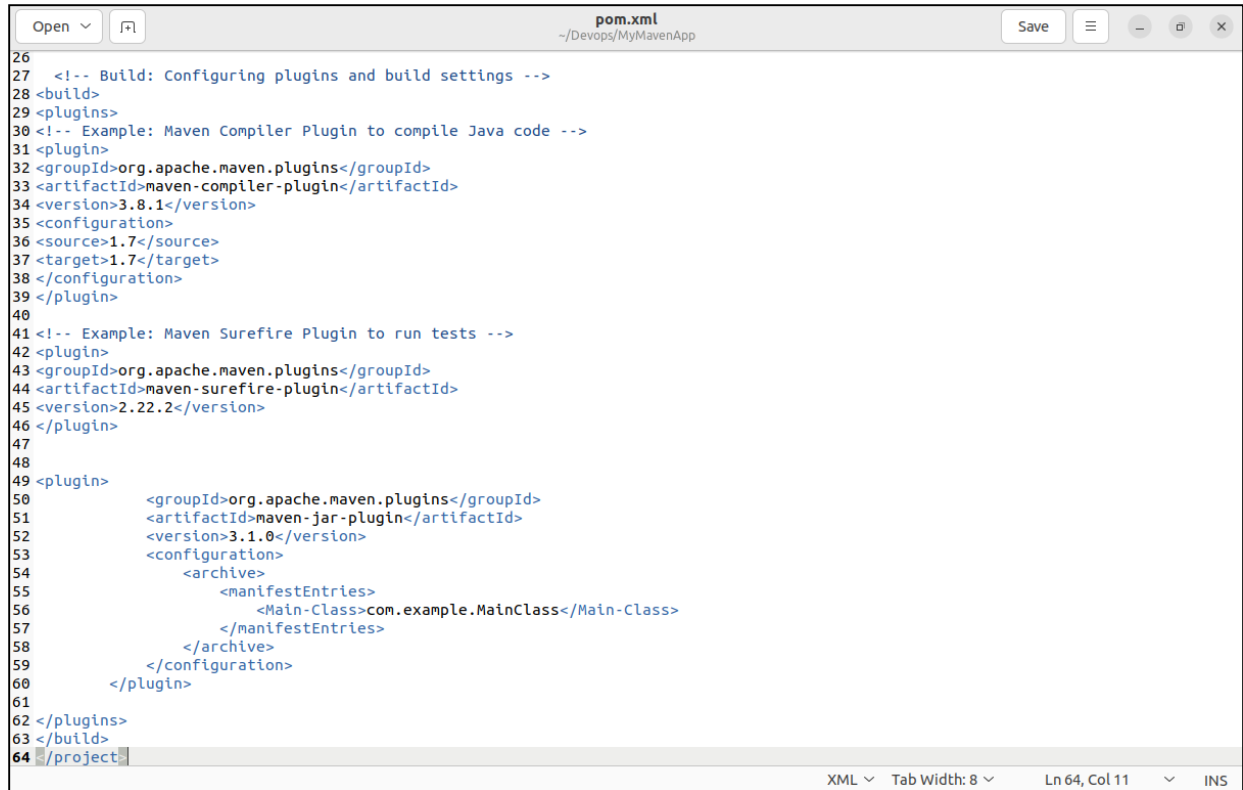
The main error is “Source option 5 is no longer supported” which means that pom.xml is either implicitly defaulting to Java 5 or that the Maven compiler

plugin doesn't know what version to use. Java 5 is deprecated and unsupported by modern JDKs, plus our pom.xml file is either lacking some packages or dependencies are not yet clearly defined so we can update the pom.xml code to following:

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A screenshot of an IDE window titled 'pom.xml' showing the following XML content:

```
26
27 <!-- Build: Configuring plugins and build settings -->
28 <build>
29 <plugins>
30 <!-- Example: Maven Compiler Plugin to compile Java code -->
31 <plugin>
32 <groupId>org.apache.maven.plugins</groupId>
33 <artifactId>maven-compiler-plugin</artifactId>
34 <version>3.8.1</version>
35 <configuration>
36 <source>1.7</source>
37 <target>1.7</target>
38 </configuration>
39 </plugin>
40
41 <!-- Example: Maven Surefire Plugin to run tests -->
42 <plugin>
43 <groupId>org.apache.maven.plugins</groupId>
44 <artifactId>maven-surefire-plugin</artifactId>
45 <version>2.22.2</version>
46 </plugin>
47
48
49 <plugin>
50     <groupId>org.apache.maven.plugins</groupId>
51     <artifactId>maven-jar-plugin</artifactId>
52     <version>3.1.0</version>
53     <configuration>
54         <archive>
55             <manifestEntries>
56                 <Main-Class>com.example.MainClass</Main-Class>
57             </manifestEntries>
58         </archive>
59     </configuration>
60 </plugin>
61
62 </plugins>
63 </build>
64 </project>
```

The IDE interface includes a top bar with 'Open', 'Save', and window controls. The bottom status bar shows 'XML', 'Tab Width: 8', 'Ln 64, Col 11', and 'INS'.

Changes made to original pom.xml

- i. Java version is now clearly specified as **Java 11** in <properties>.
- ii. Explicit <build> section with **maven-compiler-plugin** has been added, ensuring consistent Java version at build time.
- iii. **Surefire plugin** is added for running unit tests which enables better test handling (e.g., JUnit tests).
- iv. **maven-jar-plugin** is added with manifest configuration to allow running the application as an executable JAR file.

Step 7: Use **mvn clean install** command again to see the results

```
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenApp ---
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0/plexus-utils-3.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0/plexus-utils-3.0.pom (4.1 k
6 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/spice/spice-parent/16/spice-parent-16.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/spice/spice-parent/16/spice-parent-16.pom (8.4 kB a
3/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/forge/forge-parent/5/forge-parent-5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/forge/forge-parent/5/forge-parent-5.pom (8.4 kB at
5/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0/plexus-utils-3.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0/plexus-utils-3.0.jar (226 k
26 kB/s)
[INFO] Deleting /home/shweta/Devops/MyMavenApp/target
[INFO]
[INFO] at 32 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.5/plexus-utils-3.0.5.jar (2
at 188 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar (12
9.6 kB/s)
[INFO] Installing /home/shweta/Devops/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/shweta/.m2/repository/com/example
enApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/shweta/Devops/MyMavenApp/pom.xml to /home/shweta/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/My
pp-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 14.009 s
[INFO] Finished at: 2026-02-25T14:26:46+05:30
[INFO] -----
```

Step 8: Run the following command to run the JAR file:

java -jar target/MyMavenApp-1.0-SNAPSHOT.jar

```
[INFO] -----
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ java -jar target/MyMavenApp-1.0-SNAPSHOT.jar
Hello I hope everyone is doing great
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$
```


Pushing MyMavenApp to Github:

```
shweta@shweta-VirtualBox: ~/Devops/MyMavenApp

[INFO] -----
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ java -jar target/MyMavenApp-1.0-SNAPSHOT.jar
Hello I hope everyone is doing great
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ cd ..
shweta@shweta-VirtualBox:~/Devops$ gedit MyToken
shweta@shweta-VirtualBox:~/Devops$ cd MyMavenApp
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git config --global user.name "Shweta311204"
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git config --global user.email "shwetakumari311204@gmail.com"
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/shweta/Devops/MyMavenApp/.git/
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git add .
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git commit -m "initial commit"
[master (root-commit) 3356dc4] initial commit
15 files changed, 197 insertions(+)
create mode 100644 pom.xml
create mode 100644 src/main/java/com/example/App.java
create mode 100644 src/main/java/com/example/MainClass.java
create mode 100644 src/test/java/com/example/AppTest.java
create mode 100644 target/MyMavenApp-1.0-SNAPSHOT.jar
create mode 100644 target/classes/com/example/App.class
create mode 100644 target/classes/com/example/MainClass.class
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
create mode 100644 target/surefire-reports/com.example.AppTest.txt
create mode 100644 target/test-classes/com/example/AppTest.class
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$

shweta@shweta-VirtualBox: ~/Devops/MyMavenApp

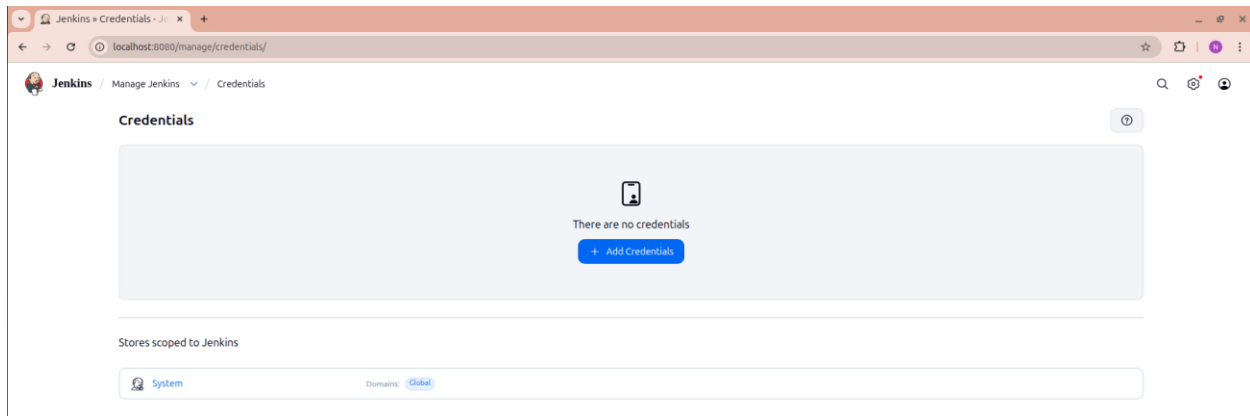
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
create mode 100644 target/surefire-reports/com.example.AppTest.txt
create mode 100644 target/test-classes/com/example/AppTest.class
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git remote aaadd origin https://github.com/Shweta311204/MyMavenApp.git
error: Unknown subcommand: aaadd
usage: git remote [-v | --verbose]
or: git remote add [-t <branch>] [-m <master>] [-f] [--tags | --no-tags] [--mirror=<fetch|push>] <name> <url>
or: git remote rename <old> <new>
or: git remote remove <name>
or: git remote set-head <name> (-a | --auto | -d | --delete | <branch>)
or: git remote [-v | --verbose] show [-n] <name>
or: git remote prune [-n | --dry-run] <name>
or: git remote [-v | --verbose] update [-p | --prune] [(<group> | <remote>)...]
or: git remote set-branches [--add] <name> <branch>...
or: git remote get-url [--push] [--all] <name>
or: git remote set-url [--push] <name> <newurl> [<oldurl>]
or: git remote set-url --add <name> <newurl>
or: git remote set-url --delete <name> <url>

-v, --verbose          be verbose; must be placed before a subcommand

shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git remote add origin https://github.com/Shweta311204/MyMavenApp.git
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git remote set-url origin https://ghp_iQyXF5wI04VOZLR8oBXoMcWt74eids0mW4p3@github.com/Shweta311204/MyMavenApp.git
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$ git push -u origin master
Enumerating objects: 41, done.
Counting objects: 100% (41/41), done.
Delta compression using up to 3 threads
Compressing objects: 100% (24/24), done.
Writing objects: 100% (41/41), 7.52 KiB | 1.50 MiB/s, done.
Total 41 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Shweta311204/MyMavenApp.git
 * [new branch]      master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
shweta@shweta-VirtualBox:~/Devops/MyMavenApp$
```

Continuous Integration (CI) Pipeline with Jenkins & Maven

Step 1: Login to Jenkins and navigate to the Credentials page, where you can manage secure information for your CI pipeline. The page shows that no credentials are currently added, and a “+ Add Credentials” button is available to create new ones. Under “Stores Scoped to Jenkins,” the **System** → **Global** domain is visible, which is used to store credentials for pipeline access to external services like GitHub or Docker.



Step 2: Add new credentials in Jenkins for your CI pipeline, where you first choose the credential type such as “Username with password” for Git or API access. Then, enter details like username, password, set the scope to Global, and provide a description (e.g., GitHub Credential). Finally, click “Create” to securely store and use the credential in Jenkins pipelines.

Add Credentials

×

Select a type of credential

Username with password
Commonly used for authentication to services like Git, APIs, or registries.

GitHub App

SSH Username with private key

Secret file

Secret text

Certificate
Upload a PKCS#12 or PEM encoded certificate and private key.

Next

Add Username with password

×

Scope ?
Global (Jenkins, nodes, items, all child items, etc)

Username
Shweta311204

☒ Treat username as secret ?

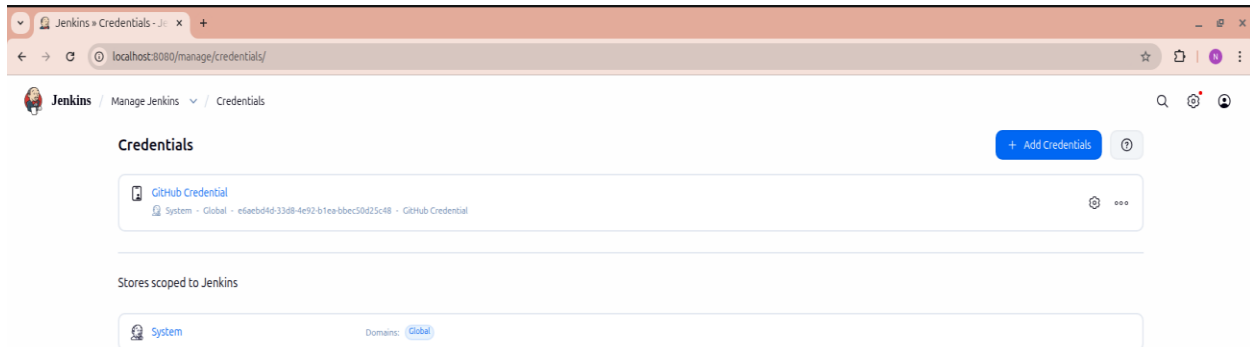
Password

ID ?

Description ?

Create

The credential has been successfully created and is now available for use in your Jenkins pipeline.



Step 3: To deploy **MyMavenApp**, open the terminal, cd into the **MyMavenApp** folder, create a **Jenkinsfile** containing the pipeline code, and then commit and push it to GitHub.

```
pipeline {
  agent any // Use any available agent
```

```
tools {
```

```
  maven 'Maven' // Ensure this matches the name configured in Jenkins
}
```

```
stages {
```

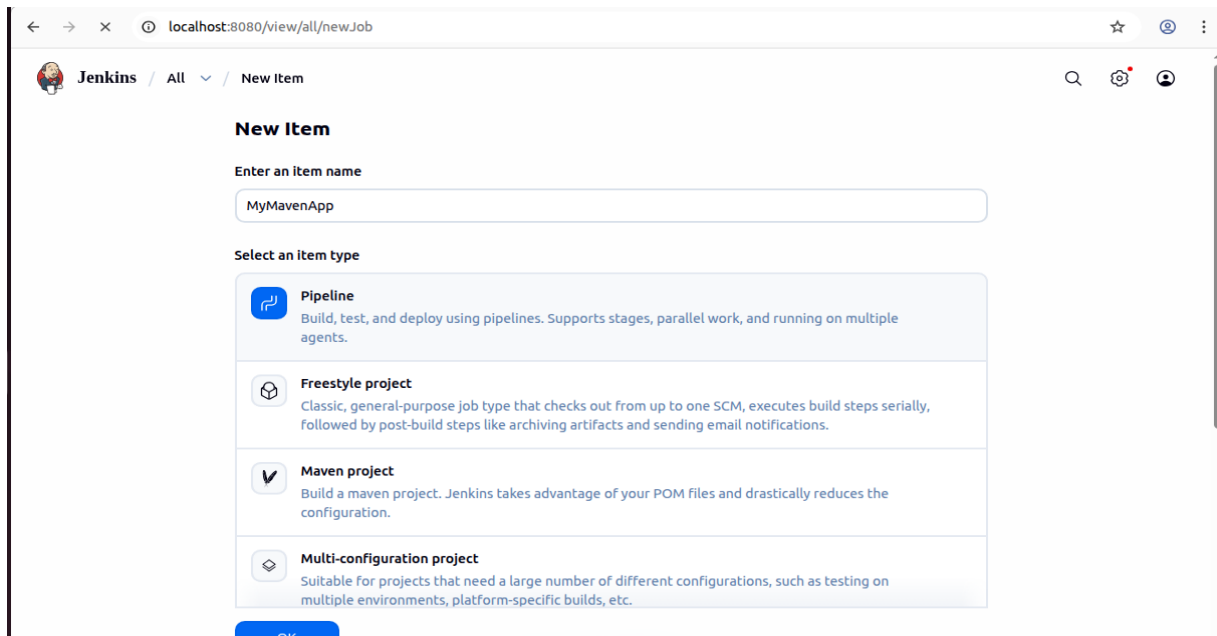
```
  stage('Checkout') { steps {
```

```
    git branch: 'master', url: 'https://github.com/Shweta311204/MyMavenApp.git'
  }
}
```

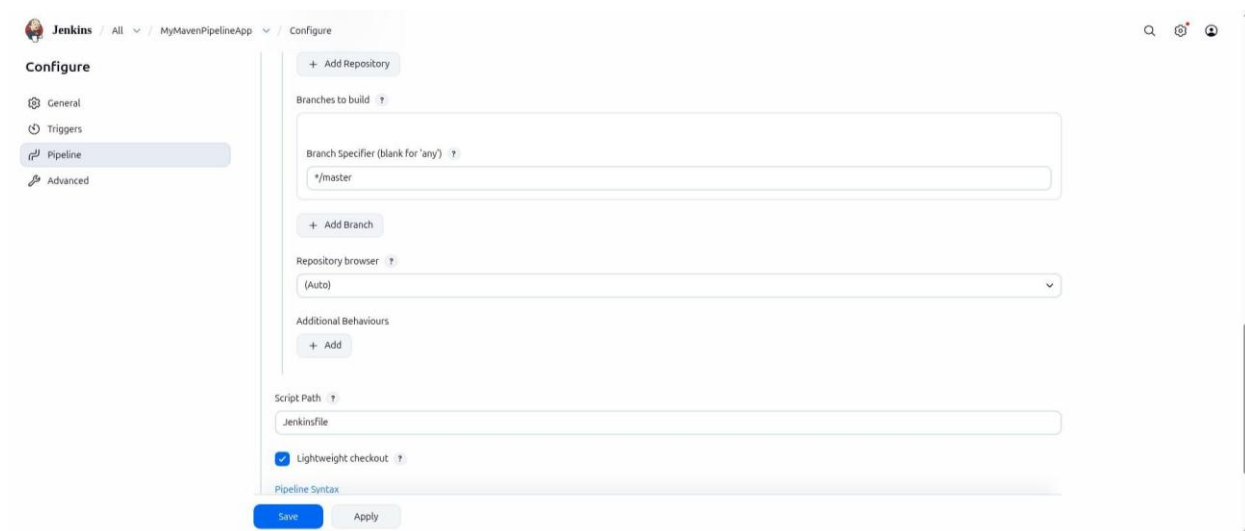
```
  stage('Build') { steps {
    sh 'mvn clean package' // Run Maven build
```

```
}  
}  
  
stage('Test') { steps {  
sh 'mvn test' // Run unit tests  
}  
}  
  
stage('Run Application') { steps {  
  
// Start the JAR application  
  
sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'  
}  
}  
}  
  
post {  
  
success {  
  
echo 'Build and deployment successful!'  
}  
  
failure {  
  
echo 'Build failed!'  
}  
}  
}
```

Step 4: To deploy **MyMavenApp**, you first need to create a **Jenkins pipeline job** that will use the Jenkinsfile from your project repository. Then, **Go to the Jenkins home page and click on “New Item” to create a job**, then enter the item name (e.g., *MyMavenApp*) and select **Pipeline** as the project type. This option allows you to build, test, and deploy applications using a Jenkinsfile. After selecting the pipeline type, click **“OK”** to proceed to the job configuration page.



Step 5: In **MyMavenApp**, go to the **Pipeline** section and under **Definition**, select **Pipeline script from SCM**. Choose **Git** as the SCM, and under **Repository URL**, paste the GitHub URL of **MyMavenApp**. In the **Credentials** field, select the previously created credential (e.g., *GitHub Credential*). Finally, in **Script Path**, enter the name of the Jenkinsfile you created in the MyMavenApp folder and pushed to GitHub. Then click **Apply** and **Save** to finalize the configuration.

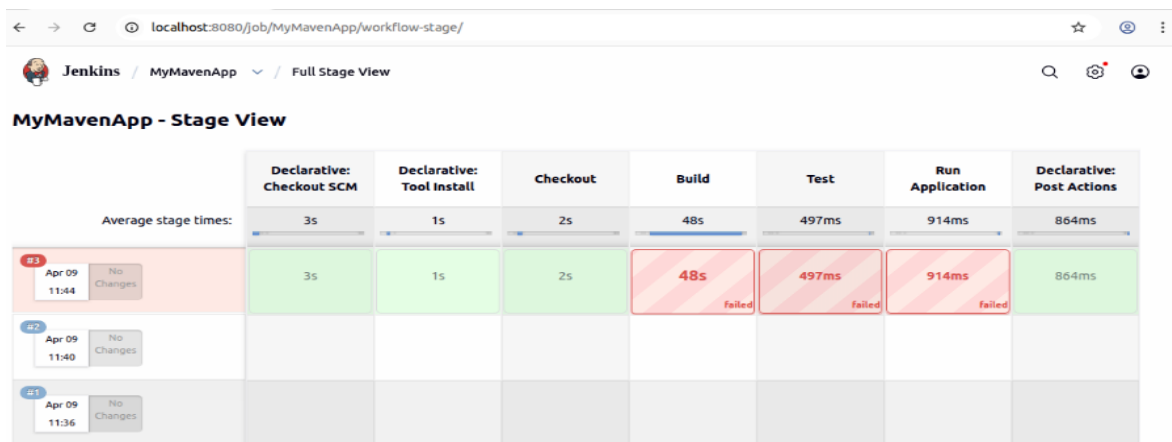


Step 6: After clicking **Apply** and **Save**, go back to **MyMavenApp** in Jenkins and click **Build Now**. If the build succeeds, there is no issue. However, if errors appear under **Build**, **Test**, or **Run Application**, open your GitHub account, go to the **MyMavenApp** repository, and edit the pom.xml file to update the Java for version (example, change 1.7 to 11). Save the changes, then return to Jenkins and click **Build Now** again; the project should now build successfully.

```

Code   Blame   64 lines (56 loc) · 1.8 KB
20     <groupId>junit</groupId>
21     <artifactId>junit</artifactId>
22     <version>3.8.1</version>
23     <scope>test</scope>
24     </dependency>
25 </dependencies>
26
27     <!-- Build: Configuring plugins and build settings -->
28     <build>
29     <plugins>
30     <!-- Example: Maven Compiler Plugin to compile Java code -->
31     <plugin>
32     <groupId>org.apache.maven.plugins</groupId>
33     <artifactId>maven-compiler-plugin</artifactId>
34     <version>3.8.1</version>
35     <configuration>
36     <source>11</source>
37     <target>11</target>
38     </configuration>
39     </plugin>
40

```



Jenkins 2.558

After updating the Java version to 11 in the pom.xml, the pipeline build ran successfully without any errors, as shown below.

